

# Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

**Input Ontology definition:** Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

MILU's 8 macro-domain / 41-subject taxonomy [Q42] aligns partially with central-exam priority subjects: Social Sciences and Law & Governance map to History/Polity, while Arts & Humanities covers cultural GK. However, the taxonomy does not separately enumerate Mathematics/Reasoning or Current Affairs as named categories — both heavily weighted in UPSC CSAT and SSC CGL Tier 1 [WEB-9, WEB-10, WEB-15]. Dataset analysis confirms reasoning and quantitative items exist (in the English partition: D8, D9), but the Hindi sample shows a heavy skew toward translated Engineering & Tech items (DC motor, half-wave rectifier, transformer, AM bandwidth, Fortran 77) that are peripheral to UPSC GS / SSC GA syllabi. State-level and central-level items are pooled without metadata distinguishing them [Q27], making it impossible to filter for central-exam scope [D24, D27].

| ID     | Evidence                                                                                                                                                                                                                                                                                                |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27    | "Our benchmark includes questions from over 40 different types of exams conducted both at the national and state levels over recent years." (p.3)                                                                                                                                                       |
| Q42    | "Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4) |
| D8     | Fill in the gap below with suitable word. Fruit : Reap:: Flower : ____... Bloom — Logical Reasoning/analogy question in English — confirms presence of reasoning subject                                                                                                                                |
| D9     | If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest ... what is the annual rate of interest? ... 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition                                                                          |
| D24    | Which one of the following statements about Legislative Committee system of Maharashtra is not correct? ... Estimate Committee: All members of this committee are from Assembly only. — State-level (Maharashtra) governance question in English partition — not central exam scope                     |
| D27    | मध्यप्रदेश में निम्नलिखित में से कौन सा उद्योग है? ... मारवा — State-level GK (Madhya Pradesh industrial geography) in Telugu — regional, not central exam                                                                                                                                              |
| WEB-9  | Current Affairs is heavily and variably weighted in UPSC GS Paper 1, raising staleness concerns for older exam scrapes (https://testbook.com/ias-preparation/upsc-prelims-subject-wise-weightage)                                                                                                       |
| WEB-10 | SSC CGL Tier 1 covers Reasoning, General Awareness, Quantitative Aptitude, English Comprehension (https://www.pw.live/ssc/exams/ssc-cgl-syllabus)                                                                                                                                                       |
| WEB-15 | UPSC CSAT Paper 2 covers Quantitative Aptitude and Logical Reasoning (qualifying) (https://vajiramandravi.com/upsc-exam/csatsyllabus/)                                                                                                                                                                  |

## Strengths

- Explicit India-first design covering Science, Social Sciences, Humanities, Arts, Business Studies, Law [Q23, Q74] aligns with multi-domain GK breadth required for UPSC/SSC.
- Hierarchical 8-domain × 41-subject structure [Q42] supports per-subject performance analysis relevant to syllabus mapping.
- Reasoning and quantitative aptitude content is present in MILU's taxonomy, confirmed empirically in the English partition [DATASET-D8, D9].

| ID | Evidence |
|----|----------|
|----|----------|

|            |                                                                                                                                                                                                                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q23</b> | “This benchmark covers many domains, including Science, Social Sciences, Humanities, Arts, Business Studies, and Law, among others.” (p.3)                                                                                                                                                              |
| <b>Q42</b> | “Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4) |
| <b>Q74</b> | “In this paper, we introduced MILU—Multilingual Indic Language Understanding Benchmark—a comprehensive benchmark specifically designed to evaluate LLMs across 11 Indic languages, spanning diverse domains and culturally relevant subjects.” (p.8)                                                    |
| <b>D8</b>  | Fill in the gap below with suitable word. Fruit : Reap:: Flower : _____ Bloom — Logical Reasoning/analogy question in English — confirms presence of reasoning subject                                                                                                                                  |
| <b>D9</b>  | If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest ... what is the annual rate of interest? ... 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition                                                                          |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                            |
|-------------|----------------------------------------------------------------|
| <b>IO-1</b> | Identify test case categories required for regional deployment |
| <b>IO-2</b> | Check if taxonomy omits regionally relevant categories         |
| <b>IO-3</b> | Check if taxonomy includes irrelevant categories               |
| <b>IO-4</b> | Document category gaps harming content validity                |

### Checklist responses:

**IO-1:** For central-exam aspirants (UPSC/SSC/banking), priority categories are GK, Current Affairs, History, Polity, Mathematics/Reasoning, Hindi proficiency, Geography, Economics (elicitation Q1; [WEB-9, WEB-10, WEB-15]).

**IO-2:** Mathematics/Reasoning and Current Affairs are not named as MILU macro-domains [Q42]; they may be subsumed under other domains but lack explicit enumeration. Dataset findings confirm reasoning/quant items exist in the English partition [DATASET-D8, D9] but not whether the Hindi partition has comparable representation. State-level vs. central-exam labeling is absent [DATASET-D24, D27].

**IO-3:** MILU pools state-PSC and central-exam items [Q27, Q92] and the observed Hindi sample contains low-priority Engineering/Tech and niche CS items (Fortran 77, AM bandwidth) that are construct-irrelevant for UPSC/SSC GS [DATASET-D5, D6, D7, D16, D36].

**IO-4:** Two notable gaps: (a) absence of explicit Current Affairs and Mathematics/Reasoning macro-domains hampers syllabus-aligned scoring; (b) inability to separate central-exam from state-PSC items at the metadata level injects construct-irrelevant variance for this deployment.

| ID         | Evidence                                                                                                                                                                                                                                                                                                |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1</b>  | “MILU spans 8 domains and 41 subjects across 11 Indic languages, reflecting both general and culturally specific knowledge.” (p.1)                                                                                                                                                                      |
| <b>Q27</b> | “Our benchmark includes questions from over 40 different types of exams conducted both at the national and state levels over recent years.” (p.3)                                                                                                                                                       |
| <b>Q42</b> | “Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4) |
| <b>Q92</b> | “We collected our questions from over 40 exam types ranging from various National and state level civil service examinations to examinations conducted by various government and private organizations.” (p.17)                                                                                         |
| <b>D5</b>  | जब एक डीसी सीरीज मोटर बना लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering/Physics question in Hindi, `is_translated: True` — not in UPSC/SSC core syllabus                                                                                                                              |
| <b>D6</b>  | हाफ वेव रेक्टिफायर का आउटपुट क्या होता है: ... प्लसेटिंग डीसी — Technical engineering question in Hindi, `is_translated: True` — domain not central to UPSC/SSC GS                                                                                                                                      |
| <b>D7</b>  | स्थिर धारा ट्रांसफार्मर _____ प्रकार का होता है। ... शेल — Electrical engineering, translated — technical domain not prioritized for Hindi-medium UPSC prep                                                                                                                                             |

|        |                                                                                                                                                                                                                                                                                            |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D8     | Fill in the gap below with suitable word. Fruit : Reap:: Flower : ____... Bloom — Logical Reasoning/analogy question in English — confirms presence of reasoning subject                                                                                                                   |
| D9     | If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest ... what is the annual rate of interest? ... 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition                                                             |
| D16    | फॉरट्रान 77 के फकिस्ड फॉरमेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? ... एक जंप लेबल या फॉरमेट लेबल — Computer Science / Fortran programming question in Hindi — very niche for UPSC/SSC aspirants                                                                            |
| D24    | Which one of the following statements about Legislative Committee system of Maharashtra is not correct? ... Estimate Committee: All members of this committee are from Assembly only. — State-level (Maharashtra) governance question in English partition — not central exam scope        |
| D27    | మధ్యప్రదేశ్‌లోని బన్‌మూర్ హరికాండ అభివృద్ధి కేంద్రం ఏ జిల్లాలో ఉంది? ... మౌరసా — State-level GK (Madhya Pradesh industrial geography) in Telugu — regional, not central exam                                                                                                               |
| D36    | एएम की बैंडविड्थ _____ है। ... 1110 kHz — AM bandwidth engineering question — highly technical, 'is_translated: True'; unlikely in UPSC GS                                                                                                                                                 |
| WEB-9  | Current Affairs is heavily and variably weighted in UPSC GS Paper 1, raising staleness concerns for older exam scrapes ( <a href="https://testbook.com/ias-preparation/upsc-prelims-subject-wise-weightage">https://testbook.com/ias-preparation/upsc-prelims-subject-wise-weightage</a> ) |
| WEB-10 | SSC CGL Tier 1 covers Reasoning, General Awareness, Quantitative Aptitude, English Comprehension ( <a href="https://www.pw.live/ssc/exams/ssc-cgl-syllabus">https://www.pw.live/ssc/exams/ssc-cgl-syllabus</a> )                                                                           |
| WEB-15 | UPSC CSAT Paper 2 covers Quantitative Aptitude and Logical Reasoning (qualifying) ( <a href="https://vajiramandravi.com/upsc-exam/csatsyllabus/">https://vajiramandravi.com/upsc-exam/csatsyllabus/</a> )                                                                                  |

## Information Gaps

- Proportion of MILU Hindi items aligned to UPSC/SSC/banking vs. state PSC is not published.
- Whether Mathematics/Reasoning and Current Affairs are present in the Hindi partition at central-exam syllabus weight is not quantifiable from the small validation sample.

## Requires Expert Verification

- A subject-matter expert in UPSC/SSC syllabus design should map MILU's 41 subjects to central-exam syllabus weights and quantify representation in the Hindi partition's test split.

## Remediation

**Gap 1: Mathematics/Reasoning and Current Affairs are not separately enumerated in MILU's 8 macro-domains; central-exam vs. state-PSC items are commingled without metadata.**

**Recommendation:** Add metadata layer mapping each MILU item to (a) UPSC/SSC/banking syllabus category (including explicit Math/Reasoning, Current Affairs), (b) central-exam vs. state-PSC scope. This enables filtered evaluation aligned to syllabus weights [WEB-9, WEB-10, WEB-15].

| ID     | Evidence                                                                                                                                                                                                                                                                                   |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WEB-9  | Current Affairs is heavily and variably weighted in UPSC GS Paper 1, raising staleness concerns for older exam scrapes ( <a href="https://testbook.com/ias-preparation/upsc-prelims-subject-wise-weightage">https://testbook.com/ias-preparation/upsc-prelims-subject-wise-weightage</a> ) |
| WEB-10 | SSC CGL Tier 1 covers Reasoning, General Awareness, Quantitative Aptitude, English Comprehension ( <a href="https://www.pw.live/ssc/exams/ssc-cgl-syllabus">https://www.pw.live/ssc/exams/ssc-cgl-syllabus</a> )                                                                           |
| WEB-15 | UPSC CSAT Paper 2 covers Quantitative Aptitude and Logical Reasoning (qualifying) ( <a href="https://vajiramandravi.com/upsc-exam/csatsyllabus/">https://vajiramandravi.com/upsc-exam/csatsyllabus/</a> )                                                                                  |